

Multiple Choice (10 marks)

1. The _____ is categorized as an unknown segment of the Deep Web which has been purposely kept hidden & is inaccessible using standard web browsers.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Dark web
 - (d) Surface web
2. Which of the following is not a transport layer vulnerability?
 - (a) Mishandling of undefined, poorly defined variables
 - (b) The Vulnerability that allows "fingerprinting" & other enumeration of host information
 - (c) Overloading of transport-layer mechanisms
 - (d) Unauthorized network access
3. Which of the following statements about a remote procedure call is true?
 - (a) It is used to call procedures with addresses that are farther than 2^{16} bytes away.
 - (b) It cannot return a value.
 - (c) It cannot pass parameters by reference.
 - (d) It cannot call procedures implemented in a different language.
4. If $p(x)$ is the minimal-degree interpolating polynomial for the real-valued function $f(x)$ at the $n + 1$ distinct real numbers $x_0, \dots, x_n$ what is the maximum possible degree of $p(x)$?
 - (a) n
 - (b) $n + 1$
 - (c) $n + 2$
 - (d) $2n$
5. Which of the following instruction-set features is NOT generally considered an obstacle to aggressive pipelining of an integer unit?
 - (a) Condition codes set by every instruction
 - (b) Variable-length encoding of instructions
 - (c) Instructions requiring widely varying numbers of cycles to execute
 - (d) Several different classes (sets) of registers

True / False (10 marks)

Answer True or False

6. True or False: A ping sweep is primarily used to locate firewalls by analyzing their response to network traffic.
7. True or False: Public key encryption simplifies key exchange by allowing secure communication without needing to share a secret key in advance.
8. True or False: The language $\{ww \mid w \text{ in } (0 + 1)^*\}$ is accepted by some Turing machine, but cannot be accepted by any pushdown automaton.
9. True or False: The digest created by a hash function is commonly referred to as a Modification Detection Code (MDC).
10. True or False: In a phishing attack, attackers rely on manipulating memory to execute harmful code directly on the victim's machine.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the harmonic sum of $n-1$.
12. Write a python function to find the sum of squares of binomial co-efficients.

End of Paper